Test Scenarios Graph Phone Book (Q-Test)

These are not one-assert-per-field checks. They are the journeys that matter: can someone register, get into contacts, and search without the whole thing falling over? Case IDs point back to 01-testcases.md.

SC-01 _ First-time operator: register, sign in, add a contact

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A new person sets up the phone book and saves one contact with a note.

1. Register with sign_up. An account should be created.
2. Sign in with sign_in. The username should appear in online_users.
3. Add a contact with add_phone_user, including name, number, and explanation. It should be saved.
4. Check with get_phone_user_by_name. The name and numbers should come back.

Covers: TC-AUTH-01, TC-AUTH-04, TC-PB001, TC-SRCH-02.
In the suite: auth tests plus test_add_phone_user_with_explanation. I kept them as separate tests with shared fixtures instead of one big pytest function, same path, less flake.

SC-02 _ Sample-file happy path

This follows the same flow as samples/commands.json: register admin, add user1 (no explanation), attach a second number, edit, list, search, then delete.

1. Sign up and sign in as admin.
2. Add user1 with add_phone_user
3. Look them up with get_phone_user_by_name.
4. Add another number with add_phone_number.
5. Edit with edit_phone_user (rename and renumber).
6. List everyone with get_all_phone_users.
7. Look up by the new name and by number.
8. Remove the contact with remove_phone_user.

Covers: TC-PB-02–07, TC-SRCH-01, TC-SRCH-04, TC-ZMQ-01–02.
Automated: command-layer coverage in test_phone_book_commands.py, plus a shorter ZeroMQ batch in test_zmq_happy_path_sign_up_and_add_contact.
Manual: run the full sample file through the real client. That is the easiest way to spot the doubleencoded dump.

SC-03 _ Session end (and what the brief assumes)

Sign in, sign out, then try to change the book.

1. Call sign_in. The user should be online.
2. Call logout. They should disappear from online_users.
3. Call add_phone_user after logout. The brief says this should be denied. The code still allows it.

Covers: TC-AUTH-07, TC-AUTH-09, TC-NEG-07.

Automated: test_logout_removes_online_user and test_phone_book_operations_do_not_require_auth. That second test is intentional _ it documents the gap.

SC-04 _ Find someone by name or number

Seed two contacts (A with two numbers, B with one), then list and search.

1. Create contact A with two numbers and contact B with one.
2. Call get_all_phone_users. A's numbers should be grouped, and the explanation should be present.
3. Call get_phone_user_by_name for A. All of A's numbers should come back.
4. Call get_phone_user_by_number with B's number. B's profile should come back.

Covers: TC-SRCH-01–04.
Automated: test_get_all_phone_users_groups_numbers, test_get_phone_user_by_number, and the by-name checks that already live in the add/edit tests.

SC-05 _ Auth failure paths

These are the cases that should not succeed quietly.

1. Try sign_up with a duplicate username. Expect failure.
2. Sign in with the wrong password. Expect a message that the password is wrong.
3. Sign in as an unknown user. Expect a user-not-found style error.
4. Call logout without logging in first. It should not claim success.

Covers: TC-AUTH-02, TC-AUTH-005–06, TC-AUTH-08.
Automated: all four above. Duplicate email (TC-AUTH-03) is still manual.

Findings: the schema does not enforce unique email.

SC-06 _ Smoke over the real wire

Start the server and talk ZeroMQ the way the system under test expects: the client binds REQ, and the server connects REP.

1. Start with python3 server.py --port <bound>. The process should be up.
2. Send a small JSON batch. A reply should come back.
3. Check the shape. Expect a list of {command_name, result} objects.

Covers: TC-ZMQ-01, TC-ZMQ-05 (batch length is roughly TC-ZMQ-02).

Finding: the reply is a JSON string wrapped in JSON _ the client has to decode twice. The automated happy-path test asserts that so we do not accidentally “fix” the bug.

An unknown command (test_zmq_unknown_command_returns_error_string) comes back as an error string instead of a result list. After that, the server’s generic except re raises, and the process may just exit.

SC-07 _ Uniqueness under real use

Check that phone numbers stay unique when more than one contact is involved.

1. Create a contact with number N. That should succeed.
2. Try to create another contact with the same number N. That should be rejected.

Covers: TC-PB-09.
Automated: test_duplicate_phone_number_rejected.

Priority and status

Here is how the scenarios stack up:

- **SC-01** _ Core onboarding. Automate: yes. Status: covered, split across modules.
- **SC-02** _ Matches the sample. Automate: yes, plus a manual full file run. Status: command layer plus a short ZeroMQ batch.
- **SC-03** _ Auth promise in the brief. Automate: yes. Status: automated; documents the missing gate.
- **SC-04** _ Main read path. Automate: yes. Status: automated.
- **SC-05** _ Bad input and abuse. Automate: partial. Status: four of five automated; email uniqueness is manual.

- **SC-06** _ Transport contract. Automate: yes. Status: two ZeroMQ tests; double-encode called out.
- **SC-07** _ Data integrity. Automate: yes. Status: automated.